

Grade 6 Accelerated Unit 10 Assessment Guide

General Information	In Unit 10, students work with a variety of data sets, represented numerically and graphically, to interpret and summarize information connected to that data. Students use their understanding of measures of center and spread to select most appropriate measures and to compare two data sets. The concepts on this assessment utilize knowledge students gained in Unit 9, so it may be helpful for the teacher to note any patterns in student successes or struggles on this assessment to determine what differentiated supports students may need. Students also use proportional reasoning in the context of data with benchmark MA.7.DP.1.3. Students are allowed the use of a calculator for this assessment. Most of the questions on this assessment are not calculator dependent. Teachers should note that an assessment limit in the draft item specifications states that items that ask students to determine the mean of a data set are limited to 10 data points, and categorical data sets are limited to six categories. This restriction is likely in place so that students do not spend an extraordinary amount of time on calculations but instead focus on making sense of the measures of center and variability in the context of the problems.
Calculator Information	The questions on the unit assessment are calculator.
Total Recommended Points	The total recommended points in this guide are 100 points.

Question	1a	Benchmark(s)	MA.7.DP.1.1
	a. The measure of center that should be used to represent the typical sale price of the homes in the neighborhood is the ☐ mean ☒ median home price. The typical price for the neighborhood is <u>\$179,000</u>.	Recommended Scoring (6 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors. Consider giving students who correctly choose median but whose work has a minor error 4 points. Consider not giving students any credit for guessing median without showing any work. Note that work could be shown on the stem-and-leaf plot if the student recognizes median should be used based on the outlier.
Question	1b	Benchmark(s)	MA.7.DP.1.1
	b. The data ☒ has an outlier, ☐ is generally symmetric, so ☐ the range is ☒ the interquartile range is ☐ both the range and the interquartile range are the preferred measure(s) of variation.	Recommended Scoring (8 Total Points)	Consider giving 4 points for each correct answer choice.

Question	2a	Benchmark(s)	MA.7.DP.1.2
	a. The southern states have a median of <u>48.76</u> strikes/km ² and a mean of <u>53.19</u> strikes/km ² .	Recommended Scoring (10 Total Points)	Consider giving 5 points for each correct measure of center. Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Question	2b	Benchmark(s)	MA.7.DP.1.2
	b. The western states have a median of <u>5.46</u> strikes/km ² and a mean of <u>6.29</u> strikes/km ² .	Recommended Scoring (10 Total Points)	Consider giving 5 points for each correct measure of center. Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Question	2c	Benchmark(s)	MA.7.DP.1.2
	c. Neither data set has an outlier. Therefore, the ○ mean ○ median should represent the center and the ○ range ○ interquartile range should be used to represent variability.	Recommended Scoring (8 Total Points)	Consider giving 4 points for each correct answer choice.

Question	2d	Benchmark(s)	MA.7.DP.1.2
	d. The western states tend to have ☐ less ☐ more variability than the southern states because the ☐ range ☐ interquartile range ☐ range and the interquartile range is/are ☐ larger ☐ smaller than the ☐ range ☐ interquartile range ☐ range and the interquartile range for the southern states.	Recommended Scoring (8 Total Points)	Consider giving 4 points for correctly choosing “less” in the first answer box. Consider giving 4 points for correctly choosing all three of the rest of the answer boxes. These boxes are interconnected.
Question	3a	Benchmark(s)	MA.7.DP.1.2
	a. A car’s stopping distance on dry pavement is typically ☐ less ☐ more than the stopping distance on wet pavement. The stopping distance on dry pavement has ☐ less ☐ more variability than the stopping distance on wet pavement.	Recommended Scoring (8 Total Points)	Consider giving 4 points for each correct answer choice.

Question	3b	Benchmark(s)	MA.7.DP.1.2
b. The stopping distance of car <i>D</i> on wet pavement is which means that the the	☐ a gap ☐ a cluster ☒ an outlier ☐ mean ☒ median ☐ range ☒ interquartile range is the best measure of center and is the best measure of spread for this data.	Recommended Scoring (12 Total Points)	Consider giving 4 points for each correct answer choice.
Question	4	Benchmark(s)	MA.6.DP.1.6
The measure of center that should be used for data set <i>B</i> is the because the two additional pulse rates that were added to data set <i>A</i> to create data set <i>B</i> influenced the	☐ mean ☒ median ☒ range. ☐ interquartile range.	Recommended Scoring (4 Total Points)	Partial credit is not recommended.
Question	5	Benchmark(s)	MA.7.DP.1.3
	1250	Recommended Scoring (8 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.

Question	6a	Benchmark(s)	MA.7.DP.1.3
Middle schoolers at Sebastian Middle School and cell phone owners		Recommended Scoring (6 Total Points)	Consider giving students 3 points for each correctly named population.
Question	6b	Benchmark(s)	MA.7.DP.1.3
- Both surveys - Neither survey - The survey conducted by a cell phone company - The survey conducted at Sebastian Middle School would be a representative sample of all cell phone users.		Recommended Scoring (4 Total Points)	No partial credit is suggested.
Question	7	Benchmark(s)	MA.7.DP.1.3
2918		Recommended Scoring (8 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.